

Introduction CSS

What is CSS?

CSS stands for Cascading Style Sheet.

CSS is a plain text file to store properties.

CSS describing look and formatting properties for how content should be display on webpage.

CSS give you control to make global change and apply all webpage.

Definition of CSS :-Cascading Style Sheets (CSS) is a rule based language that applies styling to HTML elements. We write CSS rules in elements (<p>,), and modify properties of those elements such as color, background color, width, border thickness, font size, etc.

Who Creates and Maintains CSS?

CSS was invited by **Håkon Wium Lie** on **October 10, 1994** and maintained through a group of people within the W3C called the CSS Working Group.

CSS Versions

Cascading Style Sheets, level 1 (CSS1) was came out of W3C as a recommendation in December 1996. This version describes the CSS language as well as a simple visual formatting model for all the HTML tags.

CSS2 was became a W3C recommendation in May 1998 and builds on CSS1. This version adds support for media-specific style sheets e.g. printers and aural devices, downloadable fonts, element positioning and tables.

CSS3 was became a W3C recommendation in June 1999 and builds on older versions CSS. it has divided into documentations is called as Modules and here each module having new extension features defined in CSS2.

CSS3 Modules

CSS3 Modules are having old CSS specifications as well as extension features.

- Selectors
- Box Model
- Backgrounds and Borders
- Image Values and Replaced Content
- Text Effects
- 2D/3D Transformations
- Animations
- Multiple Column Layout
- User Interface

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Why Use CSS?

Use CSS to define styles for your documents, including the design, layout and variations in display for different devices and screen sizes. You can place your CSS in the <head> of a document with an embedded style sheet, or attach a separate file that defines your styles with an external style sheet. To link an external style sheet to your document, you'll simply add a link to the style sheet in the <head> of the document.

Use of CSS is the recommended way of defining how HTML pages are displayed. You should use HTML to define the basic structure (using elements such as <h1>, <p>, , etc.) and CSS to define how these elements should appear (e.g. heading should be in bold Arial font, paragraphs should be indented, etc.).

• Uses/Advantages of CSS :-

1. **CSS saves time** - You can write CSS once and then reuse same sheet in multiple HTML pages. You can define a style for each HTML element and apply it to as many Web pages as you want.
2. **Pages load faster** - If you are using CSS, you do not need to write HTML tag attributes every time. Just write one CSS rule of a tag and apply it to all the occurrences of that tag. So less code means faster download times.
3. **Easy maintenance** - To make a global change, simply change the style, and all elements in all the web pages will be updated automatically.
4. **Superior styles to HTML** - CSS has a much wider array of attributes than HTML, so you can give a far better look to your HTML page in comparison to HTML attributes.
5. **Multiple Device Compatibility** - Style sheets allow content to be optimized for more than one type of device. By using the same HTML document, different versions of a website can be presented for handheld devices such as PDAs and cell phones or for printing.
6. **Global web standards** - Now HTML attributes are being deprecated and it is being recommended to use CSS. So its a good idea to start using CSS in all the HTML pages to make them compatible to future browsers.
7. **Offline Browsing** - CSS can store web applications locally with the help of an offline cache. Using of this, we can view offline websites. The cache also ensures faster loading and better overall performance of the website.
8. **Platform Independence** - The Script offer consistent platform independence and can support latest browsers as well.

• Style Sheet Syntax :-

A CSS comprises of style rules that are interpreted by the browser and then applied to the corresponding elements in your document. A style rule is made of three parts:

Selector: A selector is an HTML tag at which style will be applied. This could be any tag like <h1> or <table> etc.

Property
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Value
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Property: A property is a type of attribute of HTML tag. Put simply, all the HTML attributes are converted into CSS properties. They could be *color* or *border* etc.

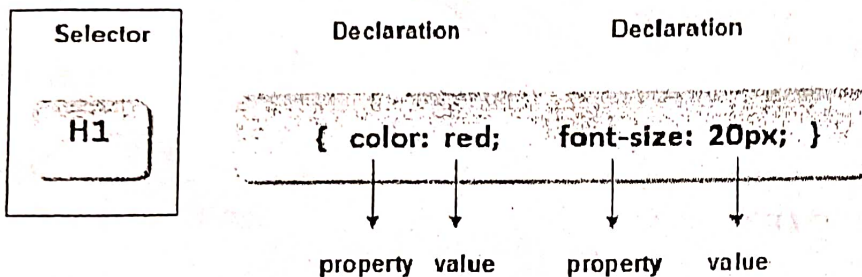
Value: Values are assigned to properties. For example *color* property can have value either *red* or *#F1F1F1* etc.

You can put CSS Style Rule Syntax as follows:

selector { property: value }

Ex :-

h1 { color:red; font-size:10px;}



- **CSS Selector :-**

Selector are used for select an Html element it is select by name, id, class etc.

1. id selector
2. class selector
3. Element Selector
4. Group Selector
5. Universal Selector

1. Element Selector :- It is used to select an Html elements by name

```
<html>
<body>
<style>
h1{
  color: red;
  font-size: 18px;
}
p{
  color: green;
  font-size: 30px; }
</style>
<h1>This is h1 heading</h1>
<p>This is paragraph</p>
</body>
</html>
```

2. id Selector :-

You can define style rules based on the id attribute of the elements. All the elements having that id will be formatted according to the defined rule. The id selector uses the id attribute of the Html element, and is defined with a "#".

```
<html>
<body>
<style>
#center
{
text-align:center;
color:red;
}
</style>
<h1 id="center">This is heading</h1>
<p id="center">This is paragraph</p>
</body>
</html>
```

3. class Selector :-

The class selector is used to specify a style for a group of elements. Unlike the id selector. This allows you to set a particular style for many Html elements with the same class. The class selector uses the Html class attribute, and is defined with a "."(dot)

```
<html>
<body>
<style>
.center
{
text-align:center;
color:green;
}
</style>
<h1 class="center">This is heading</h1>
<p class="center">This is paragraph</p>
</body>
</html>
```

You can also specify only specific Html elements should be affected by a class. In below example , all <p> elements will be display in center because here we apply css with <p> tag.

p.center {text-align:center;}

4. Group Selector :-

By using group selector you can apply same style of number of Html elements. Grouping selector is used to minimize the code. Here commas are used to separate each selector in grouping.

```
<html>
<body>
<style>
h1,p{
    color: red;
    font-size: 18px;
}
</style>
<h1>This is h1 heading</h1>
<p>This is paragraph</p>
</body>
</html>
```

5. Universal Selector

The universal selector is used to select all the elements on the pages.

```
<html>
<body>
<style>
* {
    color: red;
    font-size: 18px;
}
</style>
<h1>This is h1 heading</h1>
<p>This is paragraph</p>
</body>
</html>
```

- **CSS Comment :-**

Comment is nothing but it is a statement which is not display on browser window. It is useful to understand about particular code, which code is written for what purpose.

Syntax

```
/* css comment */
```

Ex:-

```
<html>
<body>
<style>
/* style for paragraph*/
p
{
color:red;
}
/*
multi line comment
in CSS
*/
</style>
<p>This is my first Html code</p>
</body>
</html>
```

- **Types of Style Sheet:-**

1. Inline styles
2. Internal stylesheet
3. External stylesheet

1. **Inline Styles :-** An inline style may be used to apply a unique style for a single element. To use inline styles, add the style attribute to the relevant element. The style attribute can contain any CSS property.

The following example demonstrates the inline style :-

```
<body>
<p style="font-family:arial;color:blue;font-size:20 pt"> This is a paragraph formatted
with inline css style. </p>
</body>
```


2. **Internal Style Sheet :-** An internal style sheet may be used if one single page has a unique style. Internal styles are defined within the <style> element, inside the <head> section of an HTML page:

The following example demonstrates the internal style :-

```
<html>
<head>
<style>
P {
Font-family:arial;
Font-size:medium;
Color:blue;
Text-align:justify;
}
</style>
</head>
<body>
<p> This is a paragraph formatted with internal css style. </p>
</body>
</html>
```

3. External Style Sheet

With an external style sheet, you can change the look of an entire website by changing just one file! Each page must include a reference to the external style sheet file inside the <link> element. The <link> element goes inside the <head> section:

Example

```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

An external style sheet can be written in any text editor. The file should not contain any html tags. The style sheet file must be saved with a .css extension.

Here is how the "mystyle.css" looks:

```
body {  
    background-color: lightblue;  
}  
  
h1 {  
    color: navy;  
    margin-left: 20px;  
}
```

* **The following example demonstrates the external style :-**

Create a new document in notepad, define the following css styles within it and save it with the name **my.css**

```
p  
{  
Font-family:arial;  
Font-size:medium;  
Color:blue;  
Text-align:justify;  
}  
  
h1 {  
Font-family:arial black;  
Background-color:yellow;  
Color:green;  
}
```

Create demo.html file :-

<html>

<head>


```

<link rel="stylesheet" type="text/css" href="my.css">

</head>

<body>

<h1> Heading </h1>

<p> This is paragraph formatted with External css style. </p>

</body>

</html>

```

- **Background :-**

CSS **background** property is used to define the background effects on Html element. There are 5 CSS background properties that affects the Html elements.

Property	Description
<u>background</u>	Sets all the background properties in one declaration
<u>background-attachment</u>	Sets whether a background image is fixed or scrolls with the rest of the page
<u>background-color</u>	Sets the background color of an element
<u>background-image</u>	Sets the background image for an element
<u>background-position</u>	Sets the starting position of a background image
<u>background-repeat</u>	Sets how a background image will be repeated

1. The **background-color** property is used to set the background color of HTML element.

```

<html>

<head>

<style>

body {

    background-color: lightblue;

}

h1 {

    background-color: green;

```

```

    }

    </style>

    </head>

    <body>

        <h1>Hello World!</h1>

        <p style = "background-color:yellow;">This line has a yellow background color!</p>

    </body>

</html>

```

With CSS, a color is most often specified by:

- a valid color name - like "red"
- a HEX value - like "#ff0000"
- an RGB value - like "rgb(255,0,0)"

2. The **background-image** property is used to set the background image of HTML element. By default, the image is repeated so it covers the entire element.

```

<html>

<head>

<style>

body {

    background-image: url("img/Sunset.jpg");

}

</style>

</head>

<body>

<h1>Hello World!</h1>

<p>This page has an image as the background!</p>

```



```
</body>    </html>
```

3. The **background-repeat** property is used to control the repetition of an image both horizontally and vertically in the background. You can use *no-repeat* value for *background-repeat* property if you don't want to repeat an image, in this case image will display only once.

```
<html>
<head>
<style>
body {
    background-image: url("img/Sunset.jpg");
    background-repeat: repeat-x;
}
</style>
</head>
<body>
<h1>Hello World!</h1>
<p>Here, a background image is repeated only horizontally!</p>
</body> </html>
```

To repeat an image vertically, set background-repeat: repeat-y;

4. The **background-position** property is used to control the position of an image in the background. **background-position** is use only in case of when you use image in background of any Html elements. By default, the background image is placed on the **top-left** of the webpage. You can set position of background, all background image position are given below :- **1. top 2. bottom 3. center 4. left 5. right**

2100px 200px

```
<html>
<head>
<style>
body {
    background-image: url("img/Desert.jpg");
    background-repeat: no-repeat;
    background-position: right top;
}
</style>
```

```

</head>

<body>

<h1>Hello World!</h1>

<p>W3Schools background no-repeat, set position example.</p>

<p>In this example we have also added a margin on the right side, so the
background image will never disturb the text.</p>

</body>

</html>

```

5. The **background-attachment** property is used to control the scrolling of an image in the background. Background attachment determines whether a background image is fixed(background-attachment: fixed) or scrolls with the rest of the page.(background-attachment: scroll)

```

<html>
<head>
<style>
body {
    background-image: url("img/Desert.jpg");
    background-repeat: no-repeat;
    background-position: right top;
    background-attachment: fixed;
}
</style>
</head>
<body>

<h1>Hello World!</h1>
<h2>Hello World!</h2>
<h3>Hello World!</h3>
<h4>Hello World!</h4>
<h5>Hello World!</h5>
</body>
</html>

```


• CSS Text :-

CSS TEXT properties are various type like text color, text-align, text-decoration, letter-spacing and many more properties.

Using this properties you can change the text formatting style.

Property	Description
<u>color</u>	Sets the color of text
<u>direction</u>	Specifies the text direction/writing direction
<u>letter-spacing</u>	Increases or decreases the space between characters in a text
<u>line-height</u>	Sets the line height
<u>text-align</u>	Specifies the horizontal alignment of text
<u>text-decoration</u>	Specifies the decoration added to text
<u>text-indent</u>	Specifies the indentation of the first line in a text-block
<u>text-shadow</u>	Specifies the shadow effect added to text
<u>text-transform</u>	Controls the capitalization of text
<u>text-overflow</u>	Specifies how overflowed content that is not displayed should be signaled to the user
<u>unicode-bidi</u>	Used together with the <u>direction</u> property to set or return whether the text should be overridden to support multiple languages in the same document
<u>vertical-align</u>	Sets the vertical alignment of an element
<u>white-space</u>	Specifies how white-space inside an element is handled
<u>word-spacing</u>	Increases or decreases the space between words in a text

1. The **color** property is used to set the color of a text.

<html>

<head>

```

<style>
body {
    color: blue;
}
h1 {
    color: green;
}
</style>
</head>
<body>
<h1>This is heading 1</h1>
<p>This is an ordinary paragraph. Notice that this text is blue. The default text
color for a page is defined in the body selector.</p>
</body>    </html>

```

2. The **direction** property is used to set the text direction.

```

<p style="direction:rtl;">
    This text will be rendered from right to left    </p>

```

3. The **letter-spacing** property is used to add or subtract space between the letters that make up a word.

```

<html>
<head>
<style>
h1 {
    letter-spacing: 10px;
}
h2 {
    letter-spacing: -10px;
}
</style>
</head>

```



```
<body>
<h1>This is heading 1</h1>
<h2>This is heading 2</h2>
</body>
</html>
```

4. The **word-spacing** property is used to add or subtract space between the words of a sentence.

```
<p style="word-spacing:30px;">
  This text is having space between words. </p>
```

5. The **text-indent** property is used to indent the text of a paragraph.

```
<p style="text-indent:10cm;">
  This text will have first line indented by 10cm and this line will remain at
  its actual position this is done by CSS text-indent property. </p>
```

6. The **text-align** property is used to align the text of a document. text-align possible value **center, left, right, or justify**.

```
<html>
<head>
<style>
h1 {
  text-align: center;
}
h2 {
  text-align: left;
}
h3 {
  text-align: right;
}
</style>
</head>
```

```

<body>

<h1>Heading 1 (center)</h1>

<h2>Heading 2 (left)</h2>

<h3>Heading 3 (right)</h3>

</body> </html>

```

7. The **text-decoration** property is used to underline, overline, blink and line-through text.

```

<p style="text-decoration:underline;">
  This will be underlined    </p>

<p style="text-decoration:line-through;">
  This will be striked through. </p>

<p style="text-decoration:overline;">
  This will have a over line.    </p>

<p style="text-decoration:blink;">
  This text will have blinking effect    </p>

```

8. The **text-transform** property is used to capitalize text or convert text to uppercase or lowercase letters. Possible values are *none*, *capitalize*, *uppercase*, *lowercase*.

```

<p style="text-transform:capitalize;">
  This will be capitalized    </p>

<p style="text-transform:uppercase;">
  This will be in uppercase    </p>

<p style="text-transform:lowercase;">
  This will be in lowercase    </p>

```

9. The **white-space** property is used to control the flow and formatting of text. CSS white-space possible value is **normal**, **pre**.

10. The **text-shadow** property is used to set the text shadow around a text.

```

<p style="text-shadow:4px 4px 8px blue;">

```


If your browser supports the CSS text-shadow property, this text will have a blue shadow. `</p>`
It has 4 values :- first value give space horizontally, Second value give space vertically, third value give brightness, fourth value color to shadow.

- **CSS Fonts :-**

CSS font is used for design text or font for display on web page. CSS font properties define the font family, boldness, size, and the style of a text.

You can set following font properties of an element -

1. The **font-family** property is used to change the face of a font.
2. The **font-style** property is used to make a font italic or oblique.
3. The **font-variant** property is used to create a small-caps effect.
4. The **font-weight** property is used to increase or decrease how bold or light a font appears.
5. The **font-size** property is used to increase or decrease the size of a font. Possible values could be *xx-small*, *x-small*, *small*, *medium*, *large*, *x-large*, *xx-large*, *smaller*, *larger*, *size in pixels* or *in %*. **Ex :- font-size: 40px;**
6. The **font** property is used as shorthand to specify a number of other font properties.

1. **font-family :-** In CSS, there are two types of font family names:

- **generic family** - a group of font families with a similar look (like "Serif" or "Monospace")
- **font family** - a specific font family (like "Times New Roman" or "Arial")

Generic family	Font family	Description
Serif	Times New Roman Georgia	Serif fonts have small lines at the ends on some characters
Sans-serif	Arial Verdana	"Sans" means without - these fonts do not have the lines at the ends of characters
Monospace	Courier New Lucida Console	All monospace characters have the same width

Ex :- font-family: "Times New Roman", Times, serif; OR font-family: Times New Roman;

`<p style="font-family:bookman old style;"> Welcome</p>`

2. The **font-style** property use to set font style like **normal, italic, oblique**.

Ex :- `<p style="font-style: italic"> Welcome</p>`

3. The **font-weight** property provides the functionality to specify how bold a font is. Possible values could be *normal, bold, bolder, lighter, 100, 200, 300, 400, 500, 600, 700, 800, 900*.

`<p style="font-weight:bold;">This font is bold.</p>`

`<p style="font-weight:bolder;">This font is bolder.</p>`

`<p style="font-weight:500;">This font is 500 weight.</p>`

4. The **font-variant** property specifies whether or not a text should be displayed in a small-caps font. In a small-caps font, all lowercase letters are converted to uppercase letters. However, the converted uppercase letters appears in a smaller font size than the original uppercase letters in the text.

Ex :- `<p style="font-variant: small-caps">Good Morning</p>`

5. The **font-size** property is used to increase or decrease the size of a font.

Ex:- `<p style="font-size:30px"> Good Morning </p>`

Example of Font property:-

`<head>`

`<style>`

`p { font-family: Times New Roman;`

`font-style: italic;`

`font-size: 40px;`

`font-weight: bold;`

`font-variant: small-caps;`

`}`

`</style>`

`<body>`


```
<p> Welcome !!!!!!! Maharashtra </p>
</body>
```

Font Properties Using Shorthand :-

The **font** shorthand property sets all the font properties in one declaration.

The properties that can be set, are (in order): "font-style font-variant font-weight font-size font-family"

The font-size and font-family values are required. If one of the other values are missing, the default values will be inserted, if any.

Ex :- <p style="font: italic bold 30px bookman old style">Web Design</p>

- **CSS Links :-**

Styling Links

Links can be styled with any CSS property (e.g. color, font-family, background, etc.).

In addition, links can be styled differently depending on what **state** they are in.

The four links states are:

- a:link - a normal, unvisited link
- a:visited - a link the user has visited
- a:hover - a link when the user mouses over it
- a:active - a link the moment it is clicked

1. The **color** property is used to change color of links.

```
a:link {
    color: red;
}
a:hover {
    color: green;
}
```

2. The **text-decoration** property is mostly used to remove **underlines** from links.


```
a:link {
    text-decoration: none;
}

a:active {
    text-decoration: underline;
}
```

3. The **background-color** property can be used to specify a background color for links:

```
a:link {
    background-color: yellow;
}
```

Example of Link :-

```
<!DOCTYPE html>

<html>

<head>

<style>

/* unvisited link */
a:link {

    color: red;

    text-decoration: none;

    background-color: yellow;
}

/* visited link */
a:visited {

    color: green;
}

/* mouse over link */
a:hover {
```

```

color: hotpink;
text-decoration: underline;
background-color: green;
font-size: 30px;
}

/* selected link */
a:active {
    color: blue;
    background-color: red;
}

</style>
</head>
<body>
<a href="demo5.html">This is a link</a>
</body>
</html>

```

• CSS – List :-

The CSS list properties allow you to:

- Set different list item markers for ordered lists
- Set different list item markers for unordered lists
- Set an image as the list item marker
- Add background colors to lists and list items

1. The **list-style-type** allows you to control the shape or appearance of the marker. It has values **disc**, **circle**, **decimal**, **lower-roman**, **upper-roman**, **lower-alpha**, **upper-alpha**, **none**.

Square

Ex:- <ul style="list-style-type: circle;"> <ol style="list-style-type: upper-roman;">

2. The **list-style-position** Specifies if the list-item markers should appear inside or outside the content flow. It can have one of the two values -

Value	Description
none	NA
inside	If the text goes onto a second line, the text will wrap underneath the marker. It will also appear indented to where the text would have started if the list had a value of outside.
outside	If the text goes onto a second line, the text will be aligned with the start of the first line (to the right of the bullet).

3. The **list-style-image** specifies an image for the marker rather than a bullet point or number.

Ex:- `<ol style="list-style-image:url(img/rose.png)">`

4. The **marker-offset** specifies the distance between a marker and the text in the list.

Ex:- `<ul style="list-style: inside square; marker-offset:2cm;">` **OR**

`<ol style="list-style: outside upper-alpha; marker-offset:2cm;">`

5. Styling List With Colors :-

We can also style lists with colors, to make them look a little more interesting. Anything added to the `<ol>` or `<ul>` tag, affects the entire list, while properties added to the `<li>` tag will affect the individual list items:

Example :-

`ol { background: blue; } OR <ol style="background-color:blue">`

`ol li { background: red; } OR <li style="background-color:blue">`

Example of List :-

`<head>`

`<style>`

`#test1{`

`list-style-type:circle;`

`list-style-position:inside;`

`}`

```
ul li {  
    background:yellow;  
}  
</style>  
</head>  
<body>  
    <ul id="test1">  
        <li>Maths</li>  
        <li>Physics</li>  
        <li>Chemistry</li>  
    </ul>  
    <ol style="list-style: outside upper-alpha;">  
        <li>Maths</li>  
        <li>Physics</li>  
        <li>Chemistry</li>  
    </ol>  
</body>
```

Another Example of List :-

```
<html>  
<head>  
<style>  
#test1{  
    list-style-type:circle;  
    list-style-position:outside;  
    background-color:pink;  
    padding: 20px;
```

```
    }  
    #test2{  
        list-style-type:square;  
        list-style-position:inside;  
        background-color:pink;  
        padding: 20px;  
    }  
    #test3{  
        list-style-type:lower-alpha;  
        list-style-position:outside;  
        background-color:pink;  
        padding: 20px;;  
    }  
    #test4 {  
        list-style-image: url(img/rose.png);  
    }  
    ul li {  
        background:yellow;  
        margin: 5px;  
    }  
    ol li {  
        background:yellow;  
    }  
</style>  
</head>
```



```

<body>
  <ul id="test1">
    <li>Maths</li>
    <li>Physics</li>
  </ul>
  <ul id="test2">
    <li>Maths</li>
    <li>Physics</li>
  </ul>
  <ol id="test3">
    <li>Maths</li>
    <li>Physics</li>
  </ol>
  <ul id="test4">
    <li > Maths</li>
    <li>Physics</li>
  </ul>
</body>
</html>

```

List-style Properties Using Shorthand:-

The ***list-style*** is shorthand property. It allows you to specify all the list properties into a single expression.

Ex :- <ol style="list-style: outside upper-alpha;">

• CSS – Tables :-

You can set following properties of a table –

Table Borders :-To specify table borders in CSS, use the **border** property. We specify a border for **<table>**, **<th>**, and **<td>** elements.

Ex :- `<table style="border:2px solid black;">`

Table Width and Height :-Width and height of a table are defined by the width and height properties. **Ex** :- `<table style=" width: 100%; height: 50px;">`

The **text-align** property sets the **horizontal** alignment (like left, right, or center) of the content in **<th>** or **<td>**. By default, the content of **<th>** elements are center-aligned and the content of **<td>** elements are left-aligned. **Ex** :- `<th style="text-align:left">Name</th>`

The **vertical-align** property sets the vertical alignment (like top, bottom, or middle) of the content in **<th>** or **<td>**. By default, the vertical alignment of the content in a table is middle (for both **<th>** and **<td>** elements).

The following example sets the vertical text alignment to bottom for **<td>** elements:

Ex:- `<td style=" height: 50px; vertical-align: bottom;">Sachin </td>`

Table Color:- Ex of background color and text color of **<table>** **<td>****<tr>** **<th>** elements

Ex:- `<th style=" background-color: green; color: white;">Sachin</th>`

Hoverable Table:- Use the **:hover** selector on **<tr>** to highlight table rows on mouse over:

Ex:- `tr:hover {background-color: red}`

1. The **border-collapse** property sets whether the table borders should be collapsed into a single border. This property can have two values **collapse** and **separate** (by Default).
2. The **border-spacing** specifies the width that should appear between table cells.

Ex :- `border-spacing:10px;`

3. You use the **caption-side** property to control the placement of the table caption. By default, these are rendered above the table in the document.. This property can have one of the four values **top**, **bottom**, **left** or **right**.
4. The **empty-cells** specifies whether the border should be shown if a cell is empty. This property can have one of the three values - **show**, **hide** or **inherit**.
5. The **table-layout** allows browsers to speed up layout of a table by using the first width properties it comes across for the rest of a column rather than having to load the whole table before rendering it. This property can have one of the three values: **fixed**, **auto** or **inherit**. **Ex**:- `table-layout: fixed;`

Example of Table:-

`<head>`

```

<style>
table {
    border-collapse: separate;
    caption-side: bottom;
    border-spacing: 40px;
    empty-cells: hide;
    table-layout: fixed;
}

table, td, th {
    border: 1px solid black;
    height: 80px;
    width: 15%
}

tr: hover {
    background-color: red;
    font-size: 30px;
}

</style>
</head>
<body>
<table>
<caption> Student Record</caption>
<tr style="background-color: green; color: white">
    <th>Name</th>
    <th>Salary</th>

```



```

</tr>

<tr style="text-align:right">

  <td>Peter</td>

  <td>80000000</td>

</tr>

<tr style="vertical-align:bottom">

  <td>Lois</td>

  <td></td>

</tr>

</table>

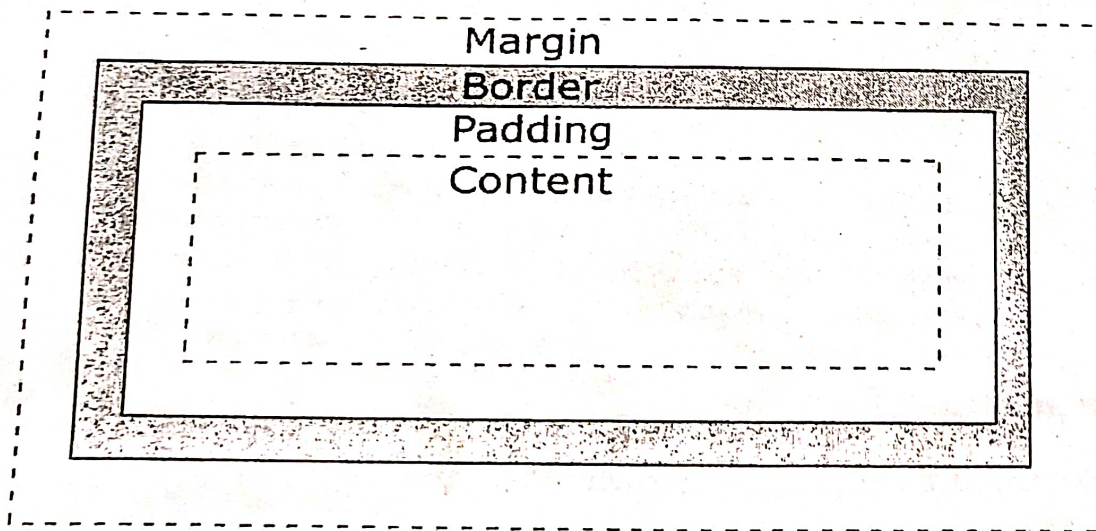
</body>

```

- **The CSS Box Model :-**

All HTML elements can be considered as boxes. In CSS, the term "box model" is used when talking about design and layout. The CSS box model is essentially a box that wraps around HTML elements, and it consists of: **margins, borders, padding, and the actual content**. The box model allows us to add a border around elements, and to define space between elements.

The image below illustrates the box model:



Explanation of the different parts:

- **Content** - The content of the box, where text and images appear
- **Padding** - Clears an area around the content. The padding is transparent
- **Border** - A border that goes around the padding and content
- **Margin** - Clears an area outside the border. The margin is transparent

Example:

```
<!DOCTYPE html>

<html>

<head>

<style>

p{

    width: 320px;

    padding: 10px;

    border: 5px solid gray;

    margin: 40px;

}

</style>

</head>

<body>

<p> The CSS box model is essentially a box that wraps around HTML elements,
and it consists of: margins, borders, padding, and the actual content.. </p>

</body>

</html>
```

• **CSS - Borders :-**

The CSS border properties allow you to specify the style, width, and color of an element's border. Following CSS border properties are used for Html elements,

1. border-style

कॅण्डस् इरोक्स सेंटर
 सांगोला कॉलेज वसतीगृहाशेजारी
 मोप्रा: अमर केदार मो.9049818463

2. border-width
3. border-color
4. border-radius

1. The **border-color** specifies the color of a border.

The border-color property allows you to change the color of the border surrounding an element. You can individually change the color of the bottom, left, top and right sides of an element's border using the properties -

- **border-bottom-color** changes the color of bottom border.
- **border-top-color** changes the color of top border.
- **border-left-color** changes the color of left border.
- **border-right-color** changes the color of right border.

```
<head>
```

```
<style>
```

```
  .example1{
```

```
    border-style: solid;
```

```
    border-bottom-color:red ;
```

```
    border-top-color: green ;
```

```
    border-left-color:blue;
```

```
    border-right-color: yellow ; /*OR border-color: red green blue yellow; */
```

```
  }
```

```
  .example2{
```

```
    border-style:solid;
```

```
    border-color: Green ;
```

```
  }
```

```
</style>
```

```
</head>
```

```
<body>
```



```
<p class="example1">
```

This example is showing all borders in different colors.

```
</p>
```

```
<p class="example2">
```

This example is showing all borders in green color only.

```
</p>
```

```
</body>
```

2. The **border-style** property specifies what kind of border to display. The following values are allowed:

- **dotted** - Defines a dotted border
- **dashed** - Defines a dashed border
- **solid** - Defines a solid border
- **double** - Defines a double border
- **groove** - Defines a 3D grooved border. The effect depends on the border-color value
- **ridge** - Defines a 3D ridged border. The effect depends on the border-color value
- **inset** - Defines a 3D inset border. The effect depends on the border-color value
- **outset** - Defines a 3D outset border. The effect depends on the border-color value
- **none** - Defines no border
- **hidden** - Defines a hidden border

You can individually change the style of the bottom, left, top, and right borders of an element using the following properties -

- **border-bottom-style** changes the style of bottom border.
- **border-top-style** changes the style of top border.
- **border-left-style** changes the style of left border.
- **border-right-style** changes the style of right border.

Ex:- 1. `<p style="border-style:dotted">Micky Mouse</p>`

(It display all border in dotted)

2. `<p style="border-style: dotted dashed solid double">Micky Mouse</p>`

OR

`_border-top-style:dotted;            border-right-style:dashed;`

`border-bottom-style:solid;        border-left-style:double;`

3. The **border-width** specifies the width of a border. The value of this property could be either a length in **px**, **pt** or **cm** or it should be set to **thin**, **medium** or **thick**.

You can individually change the width of the bottom, top, left, and right borders of an element using the following properties -

- **border-bottom-width** changes the width of bottom border.
- **border-top-width** changes the width of top border.
- **border-left-width** changes the width of left border.
- **border-right-width** changes the width of right border.

Example of border-style and border-width:-

```
<html>
<head>
<style>
.dotted {border-style: dotted;    border-width: 10px;  }
.dashed {border-style: dashed;    border-width: thin;  }
.solid {border-style: solid;       border-width: medium;}
.double {border-style: double;     border-width: thick;  }
.groove {border-style: groove; }
.ridge {border-style: ridge;  }
.inset {border-style: inset;}
.outset {border-style: outset;}
.none {border-style: none;}
.hidden {border-style: hidden;}

.mix {
    border-style: dotted dashed solid double;
    border-width: 2px 10px 4px 20px; }
</style>
```

```

</head>
<body>
<h2>The border-style Property</h2>
<p>This property specifies what kind of border to display:</p>
<p class="dotted">A dotted border.</p>
<p class="dashed">A dashed border.</p>
<p class="solid">A solid border.</p>
<p class="double">A double border.</p>
<p class="groove">A groove border.</p>
<p class="ridge">A ridge border.</p>
<p class="inset">An inset border.</p>
<p class="outset">An outset border.</p>
<p class="none">No border.</p>
<p class="hidden">A hidden border.</p>
<p class="mix">A mixed border.</p>
</body>
</html>

```

4.border-radius are used for give the radius for border, using this you can make a circle.

```

<html>
<head>
<style>
.parar1 {
    border-style: solid;

```



```
        border-radius: 20%;
        border-color: green;
    }

    .parar2 {
        height:100px;
        width:100px;
        border-style: solid;
        border-radius: 50%;
        border-color: red;
    }
</style>
</head>
<body>
<p class="parar1">This is a solid green border</p>
<p class="parar2">This is a solid red border circle</p>
</body>
</html>
```

Border Properties Using Shorthand :-

The **border** property allows you to specify color, style, and width of lines in one property .

Ex:- border:1px solid black;

```
<html>
  <head>
  </head>
  <body>
```

```
<div style="border:4px solid red;">
This example is showing shorthand property for border.
</div>
</body>
</html>
```

• CSS Margins :-

The margin are used for give sufficient space around an element (outside the border). The **margin does not have a background color**, and it is completely transparent.

We have the following properties to set an element margin.

- The **margin-bottom** specifies the bottom margin of an element. It can have a value in length, % or auto.
- The **margin-top** specifies the top margin of an element. It can have a value in length, % or auto.
- The **margin-left** specifies the left margin of an element. It can have a value in length, % or auto.
- The **margin-right** specifies the right margin of an element. It can have a value in length, % or auto.
- **Margin - Shorthand Property :-** it is possible to specify all the margin properties in one property.

The margin property is a shorthand property for the following sequence margin properties:

- margin-top
- margin-right
- margin-bottom
- margin-left

Ex :- `<p style="margin: 100px 150px 100px 80px">Micky Mouse</p>`

```
<html>
```

```
<head>
```

```
<style>
```

```
p { border:2px solid black; background-color:yellow; }
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<p style="margin-top:50px;"> Margin from top</p>
```

```

<p style="margin-bottom:200px;">Margin from bottom</p>
<p style="margin-right:100px;text-align:right;">Margin from right</p>
<p style="margin-left:200px;">Margin from left</p>
</body>
</html>

```

• CSS Padding :-

The padding clears an area around the content (inside the border) of an element. The padding is **affected by the background color of the element**. With CSS, you have full control over the padding. There are CSS properties for setting the padding for each side of an element (top, right, bottom, and left).

You can also set different values for the padding on each side of the box using the following properties -

- The **padding-bottom** specifies the bottom padding of an element. This can take a value in terms of length of %.
- The **padding-top** specifies the top padding of an element. This can take a value in terms of length of %.
- The **padding-left** specifies the left padding of an element. This can take a value in terms of length of %.
- The **padding-right** specifies the right padding of an element. This can take a value in terms of length of %.
- The **padding** serves as shorthand for the preceding properties.

```

<head>
<style>
p { border:2px solid black; background-color:yellow; }
</style>
</head>
<body>
<p style="padding-top:100px;">padding from top</p>
<p style="padding-bottom:100px;">padding from bottom</p>
<p style="padding-right:200px;text-align:right;">Padding from right</p>

```



```
<p style="padding-left:50px;">padding from left</p>
</body>
```

• CSS Display :-

CSS display is the most important property of CSS which is used to display Html elements on web page. Every Html element on the webpage is a rectangular box and the CSS display property specifies the type of box used for an Html element. The default display value for most elements is **block** or **inline**.

CSS display values

Values	Description
inline	It is used to displays an Html element in same line (like). <i>flex,</i>
none	It is used to hide the Html element.
block	It is used to displays an element as a block element (like <p>).
list-item	Let the element behave like a element. <i>list-item</i>
run-in <i>inline-block</i>	It Displays an element as either block or inline, depending on context.

```
<html>
<head>
<style>
.test { display:list-item;      }
</style>
</head>
<body>
<p style="display:none"> Welcome (Text Npt displayed)  </p>
Hello <p style="display:inline"> Welcome (Text displayed single line) </p>
<b style="display:block">This text display block value</b> Like PARAGRAPH element
<b class="test">This text display list-item value</b> Like LI element
```

```
<b class="test">This text display list-item value</b>
</body>
</html>
```

• CSS Position :-

CSS **position properties** are used for set the position of Html elements. css position properties are used for set the position of text, image, for display on web page.

CSS have following position properties;

1. Static
2. Fixed
3. Relative
4. Absolute

All CSS Position Properties

property	Description	value
bottom	It is used to set the bottom margin edge for a positioned box..	auto, length, %, inherit.
left	It sets a left margin edge for a positioned box.	auto, length, %, inherit.
right	It is used to set a right margin edge for a positioned box.	auto, length, %, inherit .
position	It is used to specify the type of positioning for an element.	absolute, fixed, relative, static, inherit
top	It is used to set a top margin edge for a positioned box.	auto, length, %, inherit

1. Static Positioning

HTML elements are positioned **static** by default.

Static positioned elements **are not affected by the top, bottom, left, and right properties**. An element with **position: static;** is not positioned in any special way. It is always positioned according to the normal flow of the page:

2. Relative Positioning

Relative positioning changes the position of the HTML element relative to where it normally appears. So "left:20" adds 20 pixels to the element's LEFT position.

You can use two values *top* and *left* along with the *position* property to move an HTML element anywhere in the HTML document.

3. Absolute Positioning

An element with **position: absolute** is positioned at the specified coordinates relative to your screen top-left corner. You can use two values *top* and *left* along with the *position* property to move an HTML element anywhere in the HTML document.

Example of Relative & absolute position :-

```
<head>
```

```
<style>
```

```
.test{
```

```
position:static;
```

```
background-color:yellow;
```

```
left:80px;
```

```
}
```

```
.test1{
```

```
position:relative;
```

```
left:80px;
```

```
top:20px;
```

```
background-color:yellow;
```

```
}
```

```
.test2{
```

```
position:absolute;
```

```
left:80px;
```

```
top:80px;
```

```
background-color:yellow;
```

```
}
```



```

</style>
</head>
<body>
<div class="test">    This div has static positioning. </div>
<div class="test1">    This div has relative positioning.    </div>
    <div class="test2">    This div has absolute positioning.    </div>
</body>

```

4. Fixed Positioning

Fixed positioning allows you to fix the position of an element to a particular spot on the page, regardless of scrolling. Specified coordinates will be relative to the browser window.

You can use two values *top* and *left* along with the *position* property to move an HTML element anywhere in the HTML document.

Positioning Text In an Image :-

How to position text over an image:

```

<head>
<style>
.topright {
    position: absolute;
    top: 80px;
    right: 16px;
    font-size: 18px;
}
img {
    width: 100%;
    height: auto;
    opacity: 0.3;

```

```

</style>
</head>
<body>
<h2>Image Text</h2>
<p>Add some text to an image in the top right corner:</p>

<p class="topright">Top Right</p>
</body>

```

- **CSS float:-**

The CSS **float** property is a positioning property. It is used to push an element to the left or right, allowing other element to wrap around it. It is generally used with images and layouts.

How it works :- Elements are floated only horizontally. So it is possible only to float elements left or right, not up or down.

- A floated element may be moved as far to the left or the right as possible.

Simply, it means that a floated element can display at extreme left or extreme right.

- The elements after the floating element will flow around it.
- The elements before the floating element will not be affected.
- If the image floated to the right, the texts flow around it, to the left and if the image floated to the left, the text flows around it, to the right.

CSS Float Properties

Property	Description	Values
clear	The clear property is used to avoid elements after the floating elements which flow around it.	left, right, both, none, inherit
float	It specifies whether the box should float or not.	left, right, none, inherit

CSS Float Property Values

Values	Description
none	It specifies that the element is not floated, and will be displayed just where it occurs in the text. this is a default value.
left	It is used to float the element to the left.
right	It is used to float the element to the right.
initial	It sets the property to its initial value.
inherit	It is used to inherit this property from its parent element.

```
<html>
<head>
<style>
img {
  float: right;
}
</style>
</head>
<body>
<p>
Float image from right</p>
</body>
</html>
```

- **CSS Opacity :-**

In CSS we can easily create transparent images very easily by using Opacity. The CSS **Opacity** property is used to specify the transparency of any Html element.

```
<head>
<style>
```



```
.trans {  
    opacity: 0.1;  
    height:100;  
    width:100;  
}  
.trans1 {  
    opacity: 0.3;  
    height:100;  
    width:100;  
}  
.trans2 {  
    opacity: 0.5;  
    height:100;  
    width:100;  
}  
.trans3 {  
    opacity:1;  
    height:100;  
    width:100;  
}
```

</style>

</head>

<body>

<h1>Original Image</h1>

<h1>Using Opacity Image </h1>


```



</body>
```

Transparent Hover Effect :-

The opacity property is often used together with the **:hover** selector to change the opacity on mouse-over:

```
<html>
<head>
<style>
img { opacity: 0.5; }
img:hover {
    opacity: 1.0;
    filter: alpha(opacity=100); /* For IE8 and earlier */
}
</style>
</head>
<body>


</body>
</html>
```